

AhamX Bodhi

Where I Becomes Infinite

Recommended Reading List

Advanced Certification Programme in Agentic and Generative AI
Prof. Sashikumaar Ganesan · IISc Bengaluru · 2026



What This Reading List Covers

Reading List at a Glance

- **3 Tier 1 Core books** that together span the full six-month curriculum — from LLM internals and implementation through visual understanding and NLP theory
- **8 Reference books** providing targeted depth for each module — from production LLMOps and RAG through scientific ML and federated learning
- **20+ online resources** including free courses, official documentation, and seminal papers aligned to every module
- All materials selected specifically for **building, training, fine-tuning, and deploying** LLMs and agentic systems — not generic deep learning



How to Use This Reading List

Tier 1 Core Books

- Read **concurrently** with the programme from Week 1
- One book is the implementation reference, one the visual companion, and one the academic theoretical foundation
- Treat them as live references throughout all six months — not one-time reads

Reference Books

- Consult **module by module** as you reach each topic
- Not intended to be read cover-to-cover
- Each maps to specific modules shown in the summary table
- Target chapters relevant to your current mini-project



Tier 1 Core Books --- The Three Foundations

Book Title	Year	Role in Programme
<i>Build a Large Language Model (From Scratch)</i>	2024	Implementation — code every component
<i>Hands-On Large Language Models</i>	2024	Visual companion — diagrams and applied code
<i>Speech and Language Processing</i> (3rd Ed., draft)	2025	Academic foundation — theory and formalism

How the Three Books Work Together

Build a Large Language Model teaches you to code it. *Hands-On Large Language Models* teaches you to see it. *Speech and Language Processing* teaches you to understand it. Together they form a complete learning system for the full six months.



Tier 1 Core --- *Build a Large Language Model (From Scratch)*

Manning, 2024 · github.com/rasbt/LLMs-from-scratch

- **Authors:** Sebastian Raschka
- The single most direct match to this programme's goals — builds a GPT-style LLM end-to-end in PyTorch: tokenizer, attention mechanism, pre-training, instruction fine-tuning, and RLHF, all from first principles with working code
- Every concept in M2, M3, M7, and M9 has a hands-on counterpart in this book
- Students who work through it systematically will have implemented the core of a production-grade LLM before the programme's midpoint
- Complete GitHub repository with all code and Jupyter notebooks included

Primary Modules

M2 (Transformers, Prompting) · M3 (LLM Architecture) · M7 (Fine-Tuning, RLHF) · M9 (Pre-Training, LLMOps)



Tier 1 Core --- *Hands-On Large Language Models*

O'Reilly, 2024 · github.com/HandsOnLLM/Hands-On-Large-Language-Models

- **Authors:** Jay Alammar & Maarten Grootendorst
- Originating from the acclaimed *Illustrated Transformer* blog series — the most visually accessible LLM book available; every concept is diagram-first, then code
- Covers embeddings, semantic search, text classification, fine-tuning, and RAG with rich visual explanations and Hugging Face code throughout
- Bridges theory and practice without sacrificing either — ideal for students who benefit from diagram-first understanding before engaging the mathematics
- Natural companion to *Build a Large Language Model*: where that book gives code, this book gives the visual intuition that makes the code meaningful

Primary Modules

M2 (Transformers, Embeddings) · M3 (LLM Architecture) · M5 (RAG, Semantic Search) · M7 (Fine-Tuning)



Tier 1 Core --- *Speech and Language Processing* (3rd Ed.)

Stanford, 2025 · Free: web.stanford.edu/~jurafsky/slp3

- **Authors:** Dan Jurafsky & James H. Martin
- The definitive NLP and LLM textbook, freely available and continuously updated — the 2025 draft covers language modelling, transformers, tokenization, BERT, instruction tuning, DPO, RLHF, ASR (Whisper), and TTS (EnCodec, VALL-E)
- Provides the rigorous academic scaffolding behind everything the programme builds; where *Build a Large Language Model* teaches students to build, this book teaches them to understand what they are building and why it works
- Key chapters: Ch. 7–9 (LLMs and transformers), Ch. 10–11 (fine-tuning), Ch. 15–16 (ASR and TTS) — mapping directly to M2 through M7

Primary Modules

M2 (Architecture) · M3 (LLMs, Tokenization) · M4 (Multimodal, ASR, TTS) · M7 (Fine-Tuning, Alignment)



Reference Books --- Module Mapping at a Glance

Book Title	Year	Primary Modules
<i>LLM Engineer's Handbook</i>	2024	M7, M9
<i>AI Engineering</i>	2025	M5, M8, M9
<i>Foundations of Large Language Models</i>	2024	M3, M7
<i>Building LLMs for Production</i>	2024	M2-M5, M7-M9
<i>RLHF Book</i>	2024	M7
<i>Reinforcement Learning: An Introduction</i>	2018	M7 (background)
<i>Physics-Informed Machine Learning</i>	2023	M6
<i>Federated Learning: Privacy and Incentive</i>	2020	M10

Reference --- Production Engineering



LLM Engineer's Handbook

- Packt, 2024 · Paul Iusztin & Maxime Labonne
- The most complete production LLM lifecycle book; covers data pipelines, RAG construction, fine-tuning with LoRA and QLoRA, DPO alignment, LLMOps monitoring, and AWS deployment
- Structured as a single end-to-end project (the “LLM Twin”) so every chapter builds on the last
- **Use for: M7** (LoRA, QLoRA, DPO) and **M9** (LLMOps, deployment)

AI Engineering

- O'Reilly, 2025 · Chip Huyen
- Definitive reference for the application and production layers of foundation models; covers prompt engineering, RAG, agentic patterns, LLM-as-a-judge evaluation, and deployment at scale
- Rewritten specifically for the foundation model era; strong on safety and responsible deployment
- **Use for: M5** (agents, RAG, MCP), **M8** (evaluation, safety), **M9** (production serving)



Reference --- Theory and Full-Stack Coverage

Foundations of Large Language Models

- Open access, 2024 · Tong Xiao & Jingbo Zhu
- University-level textbook covering pre-training objectives (CLM, MLM), prompting theory, alignment formalisms, and inference
- Clearest mathematical treatment of LLM training objectives available in open-access form
- Strong on scaling laws, in-context learning, and chain-of-thought formalisms
- **Use for:** Mathematical depth in **M3** and **M7**

Building LLMs for Production

- Towards AI, 2024 · Bouchard & Peters
- Practitioner guide covering the full LLM application stack: prompting, LangChain, LlamaIndex, advanced RAG, agents, fine-tuning (LoRA, SFT), and cloud deployment
- Broadest single-volume coverage of M2-M9; useful as a module-by-module lookup reference throughout the programme
- **Use for:** Practical lookup across **M2-M5, M7-M9**



Reference --- Alignment and Reinforcement Learning

RLHF Book

- Free at rlhfbook.com · 2024 · Nathan Lambert
- Dedicated open book on post-training alignment; covers reward modelling, PPO, DPO, and preference optimisation with mathematical rigour
- Written by a leading RLHF practitioner behind Zephyr-Beta, Tulu 2, and OLMo at AI2 and HuggingFace
- **Use for:** All RLHF and DPO content in **M7**

Reinforcement Learning: An Introduction

- Free at incompleteideas.net · MIT Press, 2018 · Sutton & Barto
- The canonical RL textbook; essential background for understanding PPO, reward modelling, and the full RLHF pipeline
- Read Chapters 1–6 before the RLHF sections of M7 if you have no prior RL background
- **Use for:** Foundational RL theory underpinning the alignment content of **M7**



Reference --- Scientific ML and Federated Learning

Physics-Informed Machine Learning

- Cambridge University Press, 2023 · Karniadakis et al.
- Primary reference for M6; covers PINNs, neural operators, surrogate modelling, uncertainty quantification, and domain knowledge integration
- No substitute exists at this depth for scientific computing with ML
- **Companion:** DeepXDE documentation at deepxde.readthedocs.io
- **Use for:** All content in **M6**

Federated Learning: Privacy and Incentives

- Springer, 2020 · Qiang Yang et al.
- Standard academic reference for M10; covers FedAvg, horizontal and vertical federated learning, differential privacy, and communication efficiency
- Applications in healthcare and finance map directly to M10 case studies
- **Companion:** Flower framework at flower.ai/docs
- **Use for:** All content in **M10**



Online Resources --- Foundations (M1--M3)

Free Online Reading and Interactive Resources

- 1. • **The Illustrated Transformer** — <https://jalammar.github.io/illustrated-transformer/>
The most-cited visual walkthrough of transformer architecture; natural companion to *Hands-On Large Language Models*
- **The Annotated Transformer** — nlp.seas.harvard.edu/annotated-transformer/
Line-by-line PyTorch implementation of "Attention Is All You Need"; supports M2 labs
- **Let's Build GPT** (YouTube, 2023) — [youtube.com/watch?v=kCc8FmEb1nY](https://www.youtube.com/watch?v=kCc8FmEb1nY)
Builds a character-level GPT from scratch in pure PyTorch; 4 hours; essential for M3
- 2. • **LLM Visualization** — <https://bbycroft.net/llm>
Interactive 3D visualization of the transformer forward pass and attention internals
- 3. • **Anthropic Prompt Engineering Guide**
docs.anthropic.com/en/docs/build-with-claude/prompt-engineering · M2



Online Resources --- Agentic AI and RAG (M5)

Documentation, Papers, and Guides

- **Building Effective Agents** — anthropic.com/research/building-effective-agents
Official Anthropic design patterns for agentic systems; required reading alongside M5
- **Model Context Protocol (MCP)** — modelcontextprotocol.io
Anthropic's specification for agent-tool interoperability; directly covered in M5
- **LangChain Conceptual Guides** — python.langchain.com/docs/concepts
Core reading for agent loops, tool use, and RAG pipeline construction
- **LlamaIndex Documentation** — docs.llamaindex.ai
RAG architecture, indexing strategies, and advanced retrieval pipeline design
- **ReAct: Synergizing Reasoning and Acting** — arxiv.org/abs/2210.03629
Foundational paper behind the ReAct agent loop; required reading for M5



Online Resources --- Fine-Tuning and RLHF (M7)

Papers, Blogs, and Documentation

- **Illustrating RLHF** — huggingface.co/blog/rlhf
Clearest accessible introduction to the full RLHF pipeline; read before M7 lab sessions
- **LoRA: Low-Rank Adaptation of Large Language Models** — arxiv.org/abs/2106.09685
Primary reference paper for parameter-efficient fine-tuning; core to M7
- **InstructGPT** — arxiv.org/abs/2203.02155
The paper that formalized RLHF for instruction following; essential M7 background
- **Hugging Face PEFT Documentation** — huggingface.co/docs/peft
Implementation guide for LoRA, QLoRA, prefix tuning, and adapter methods
- **Hugging Face TRL Documentation** — huggingface.co/docs/trl
Standard library for PPO, DPO, and SFT training workflows; used in M7 labs



Online Resources --- LLMOps and Scaling (M9)

Systems-Level Reading and Production Guides

- **How to Scale Your Model** — jax-ml.github.io/scaling-book/ **[Key Resource]**
Roofline analysis, TPU/GPU internals, transformer math, sharding strategies, training and inference optimisation at scale; 12 parts including a GPU section
- **The Ultra-Scale Playbook** — huggingface.co/spaces/nanotron/ultrascale-playbook
Distributed training: data parallelism, tensor parallelism, pipeline parallelism, ZeRO
- **Chinchilla Scaling Laws** — arxiv.org/abs/2203.15556 (2022)
Compute-optimal training; directly referenced in M9 scaling content
- **vLLM Documentation** — docs.vllm.ai
The leading open-source LLM serving framework; used in M9 deployment labs
- **LLM Course** — github.com/mlabonne/llm-course
End-to-end open-source LLM engineering: quantization, fine-tuning, and merging

Online Resources --- Safety, Scientific ML, and Federated (M6, M8, M10)



AI Safety and Scientific ML

- **HELM** —
crfm.stanford.edu/helm/latest
Holistic LLM evaluation framework;
M8
- **Anthropic Safety Research**
anthropic.com/research ·
Constitutional AI, red teaming; M8
- **LLM-as-a-Judge Paper**
arxiv.org/abs/2306.05685 · M8
- **PINNs Original Paper (2019)**
arxiv.org/abs/1711.10561 · M6
- **DeepXDE Documentation**
deepxde.readthedocs.io · M6

Federated and Distributed Learning

- **FedAvg Original Paper (2017)**
arxiv.org/abs/1602.05629 · M10
- **Flower Framework Documentation**
flower.ai/docs · M10
- **Hugging Face Deep RL Course**
huggingface.co/learn/deep-rl-course
· M7
- **OpenAI Guide to Building Agents**
platform.openai.com/docs/guides/agents
· M5



Structured Online Courses --- Complementary Resources

Free and Accessible Course Platforms

- **Hugging Face NLP Course** — huggingface.co/learn/nlp-course
Free, interactive, code-first. Transformers, tokenizers, fine-tuning, and RAG with Jupyter notebooks. Excellent companion for M2–M5 and M7
- **DeepLearning.AI Short Courses** — deeplearning.ai/short-courses
1–2 hour focused modules on RAG, agents, LangChain, and fine-tuning; courses by creators of LangChain, LlamaIndex, and Anthropic
- **Berkeley LLM Agents MOOC**
11 lectures by Berkeley RDI Center; agents, RAG, MCP, and multi-agent coordination; directly supports M5
- **fast.ai Practical Deep Learning** — fast.ai
Top-down, code-first; strong intuition builder before transformer mathematics; supports M1–M2
- **Neural Networks: Zero to Hero** (YouTube)
Builds intuition from backprop through GPT; supports M1–M3



Complete Reading Map --- All Modules

Mod.	Primary Book(s)	Key Online Resource(s)
M1- M2	<i>Build a Large Language Model; Hands-On LLMs</i>	Illustrated Transformer; Anthropic Prompt Guide
M3	<i>Build a Large Language Model; Foundations of LLMs</i>	Annotated Transformer; Let's Build GPT (YouTube)
M4	<i>Speech and Language Processing</i> (Ch. 15–16)	HuggingFace Vision docs
M5	<i>AI Engineering; Building LLMs for Production</i>	LangChain docs; MCP spec; ReAct paper
M6	<i>Physics-Informed Machine Learning</i>	DeepXDE docs; PINNs paper
M7	<i>Build a Large Language Model; LLM Engineer's Handbook; RLHF Book</i>	HF PEFT docs; LoRA paper; TRL docs
M8	<i>AI Engineering</i>	HELM; Anthropic safety research
M9	<i>LLM Engineer's Handbook</i>	<i>How to Scale Your Model</i> ; Ultra-Scale Playbook; vLLM
M10	<i>Federated Learning: Privacy and Incentive</i>	Flower framework; FedAvg paper



Recommended Reading Strategy

Phase 1: Weeks 1–10 (M1–M5)

- Start *Build a Large Language Model* Ch. 1–4 alongside M1–M2
- Read *Speech and Language Processing* Ch. 7–9 during M2–M3
- Use *Hands-On Large Language Models* as visual companion throughout M2–M3 and for RAG in M5
- Consult *AI Engineering* for agent design patterns in M5
- Keep LangChain and LlamaIndex docs open during M5 labs

Phase 2: Weeks 11–26 (M6–M10)

- Work through *Build a Large Language Model* Ch. 5–7 during M7
- Read *RLHF Book* before the RLHF lab in M7
- Read *Reinforcement Learning: An Introduction* Ch. 1–6 if new to RL
- Consult *LLM Engineer's Handbook* chapter-by-chapter through M9
- Use *Physics-Informed Machine Learning* and *Federated Learning: Privacy and Incentive* for M6 and M10

Key Takeaways



What Makes This Reading List Distinctive

- Every book is selected specifically for **building LLMs and agentic systems** — not generic deep learning or machine learning theory
- The three Tier 1 books form a complete learning system: *Build a Large Language Model* (code it), *Hands-On Large Language Models* (see it), *Speech and Language Processing* (understand it)
- Reference books are **module-specific** — target only the chapters that match your current module and mini-project; not intended for cover-to-cover reading
- Online resources are **active tools** — documentation, papers, and notebooks to keep open during labs, not just background reading
- *How to Scale Your Model* (jax-ml.github.io/scaling-book/) is the single most important systems-level online resource for M9 and anyone building at production scale

Thank You

Recommended Reading List
Advanced Certification Programme in Agentic and Generative AI
· IISc Bengaluru